

MTH 9875 The Volatility Surface: Fall 2026

Lecture 2: Stochastic volatility, rough volatility, and local volatility

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Outline of lecture

- Stochastic volatility
 - Motivation for stochastic volatility
 - Motivation for rough volatility
 - The $\mathbb{P}$ and $\mathbb{Q}$ measures
 - The market price of volatility risk and market completeness
 - Examples of stochastic volatility models
 - Solving the valuation equation
- Local volatility
 - Historical development
 - The Dupire equation
 - Local variance as a conditional expectation of instantaneous variance
 - Local volatility in terms of implied volatility
 - The BBF formula

Stochastic volatility

- Stochastic volatility models take the form

$$\frac{dS_t}{S_t} = \sqrt{V_t} dZ_t$$

with $V_t = \frac{d}{dt} \langle \log S \rangle_t$.

- Thus, the stock price process and the quadratic variation process are both assumed continuous.
 - There are no jumps!

Why stochastic volatility?

- Volatility obviously fluctuates
 - For example, recall the stock market crash of October 1987
- However, it may not be obvious what the practical benefits of going beyond Black-Scholes might be.
 - Black-Scholes is just fine for many applications.
- Stochastic volatility (SV) models are useful because they generate smiles that look like observed market smiles.
 - SV models are arbitrage-free by construction.
- Unlike alternative models that can fit the smile (such as local volatility models for example), SV models assume realistic dynamics for the underlying.
- In summary, SV models allow us to price exotic claims consistently with market prices, assuming realistic dynamics.

Stochastic volatility and business time

- SV price processes are sometimes accused of being *ad hoc*.
- On the contrary, they can be viewed as arising from Brownian motion subordinated to a random clock.
- As we saw in Lecture 1, this clock time may be identified with integrated variance (as estimated using conventional estimators).
 - Since the length of a calendar day in trading time varies, this can generate fat tails in stock returns as observed empirically.

Engineering considerations

- Hedgers who use the Black-Scholes model must continuously change the volatility assumption in order to match market prices.
 - The BS hedge ratio changes uncontrollably every time the volatility changes.
- SV models bring some order into this chaos.
- The prices of exotic options given by models based on Black-Scholes assumptions can be wildly wrong.
- SV models take the volatility smile into account when pricing exotic options.
 - SV model prices are closer to market prices than BS or local volatility model prices.

Log returns of SPX

```
In [1]: library(quantmod)

options("getSymbols.warning4.0"=FALSE)

getSymbols("^GSPC", from="1980-01-01") #Creates the time series object GSPC

ret.spx <- log(Cl(GSPC)/lag(Cl(GSPC)))
ret.spx <- ret.spx[!is.na(ret.spx)] # Remove missing values
ret.spx <- ret.spx - mean(ret.spx)
```

Loading required package: xts

Loading required package: zoo

Attaching package: 'zoo'

The following objects are masked from 'package:base':

as.Date, as.Date.numeric

Loading required package: TTR

Registered S3 method overwritten by 'quantmod':

method from
as.zoo.data.frame zoo

'GSPC'

```
In [2]: library(repr)
options(repr.plot.height=7, repr.plot.width=10, repr.plot.res=200)
```

```
In [3]: plot(ret.spx, main=NA, type="l", col="red2")
```

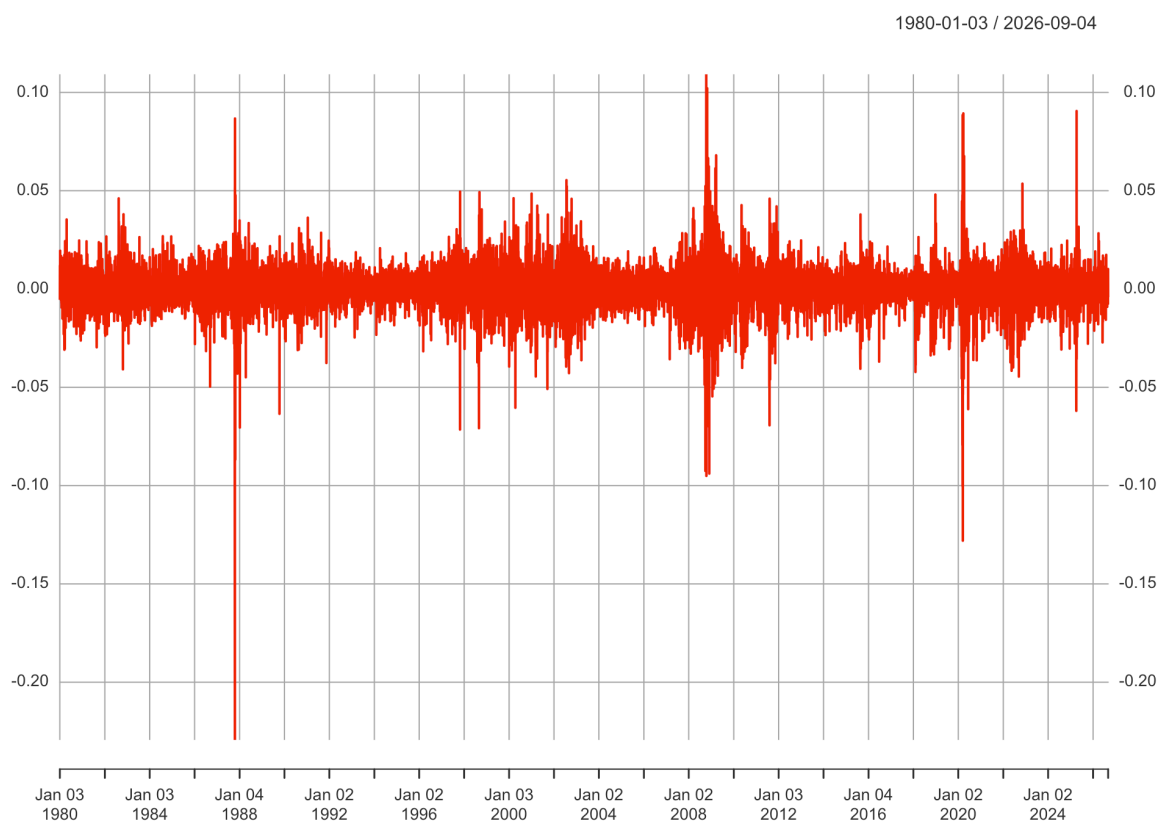


Figure1: Large moves follow large moves and small moves follow small moves: "volatility clustering".

Frequency distribution of log returns

```
In [4]: breaks <- seq(-.235, .115, .002)
spx.hist <- hist(ret.spx, breaks=breaks, plot=F)
plot(spx.hist, xlim=c(-.05, .05), freq=F, main=NA, xlab="log return")
sig <- sd(as.numeric(ret.spx))
df <- 3.2
curve(dt(x*sqrt(df)/sig, df=df)*sqrt(df)/sig, from=-.05, to=.05, col="red2", lwd=
curve(dnorm(x, mean=0, sd=sig), from=-.05, to=.05, col="royalblue", add=T)
```

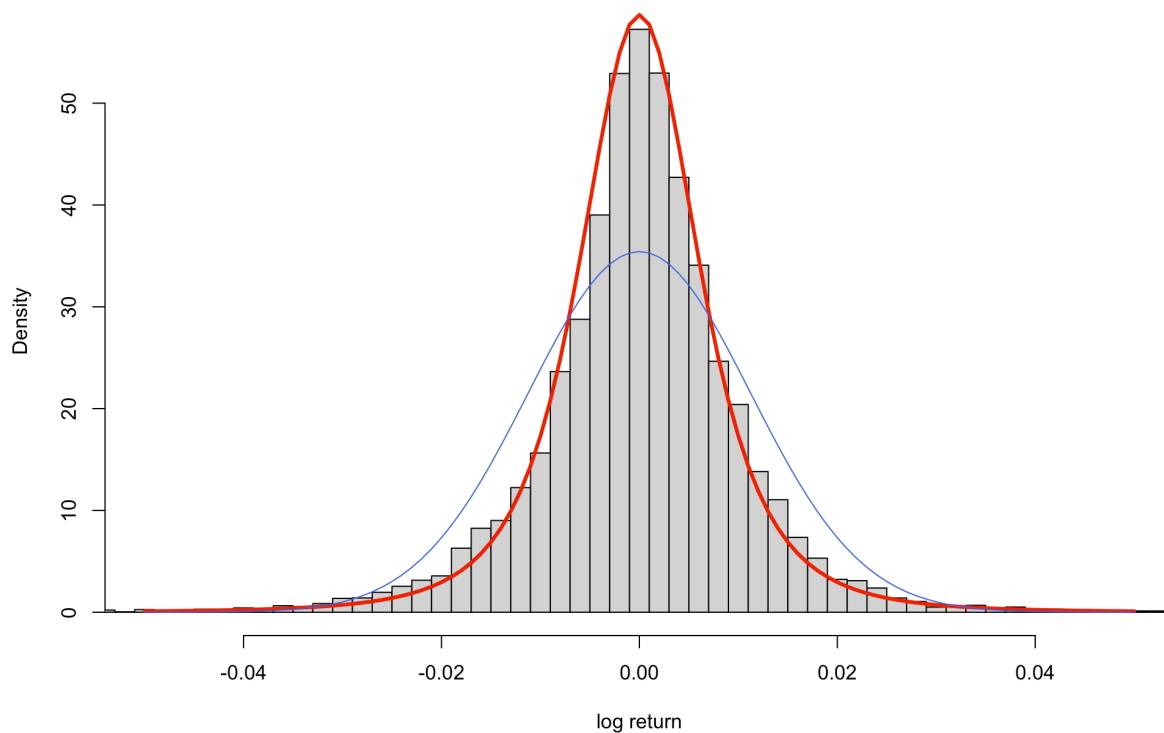


Figure 2: Frequency distribution of SPX daily log returns (since 1950) compared with the normal distribution (blue) and Student-t with 3 degrees of freedom (red).

- Note that the x-axis has been truncated; the -22.9% return on 10/19/1987 is not directly visible.

A toy regime-switching model

Suppose that realized volatility over a one-year period can be either 10% or 30%, each with probability $1/2$ (depending on an initial coin toss say).

```
In [5]: d3010 <- function(x){(dnorm(x,sd=.1)+dnorm(x,sd=.3))/2}
curve(d3010(x),from=-.8,to=.8,col="royalblue",ylim=c(0,4))
curve(dnorm(x,sd=.1),from=-.8,to=.8,col="red2",lty=2,add=T)
curve(dnorm(x,sd=.3),from=-.8,to=.8,col="green4",lty=2,add=T)
```

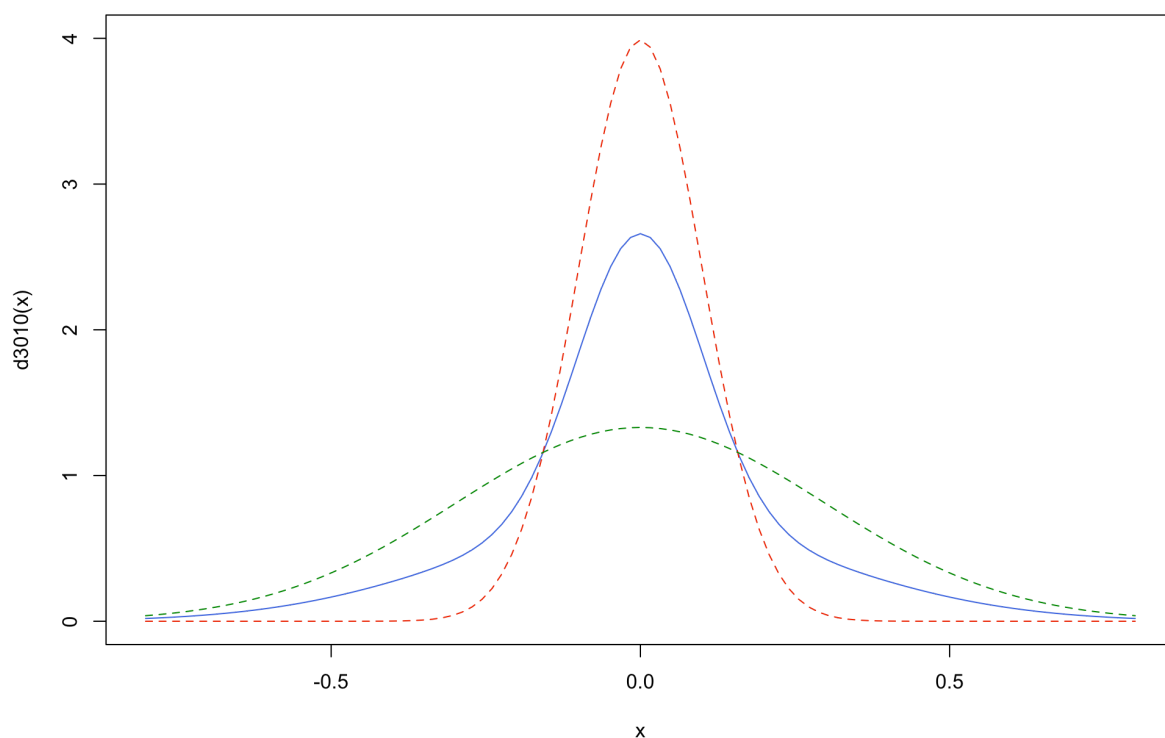


Figure 3: The two-regime density in blue; 10% and 30% volatility distributions in red and green respectively.

- In some sense, the peak of the distribution is driven by the low volatility regime and the tails by the high volatility regime.
- Fat tails and the high central peak are characteristics of mixtures of distributions with different variances.

Add more regimes

Consider the following 20-regime model:

```
In [6]: d.20regimes <- function(x){
  res <- 0
  for (i in 1:20){ res <- res + dnorm(x,sd=i/20)}
  return(res/20)
}
```

The resulting density is:

```
In [7]: curve(d.20regimes(x), from=-1.5, to=1.5, col="royalblue", n=1000)
```

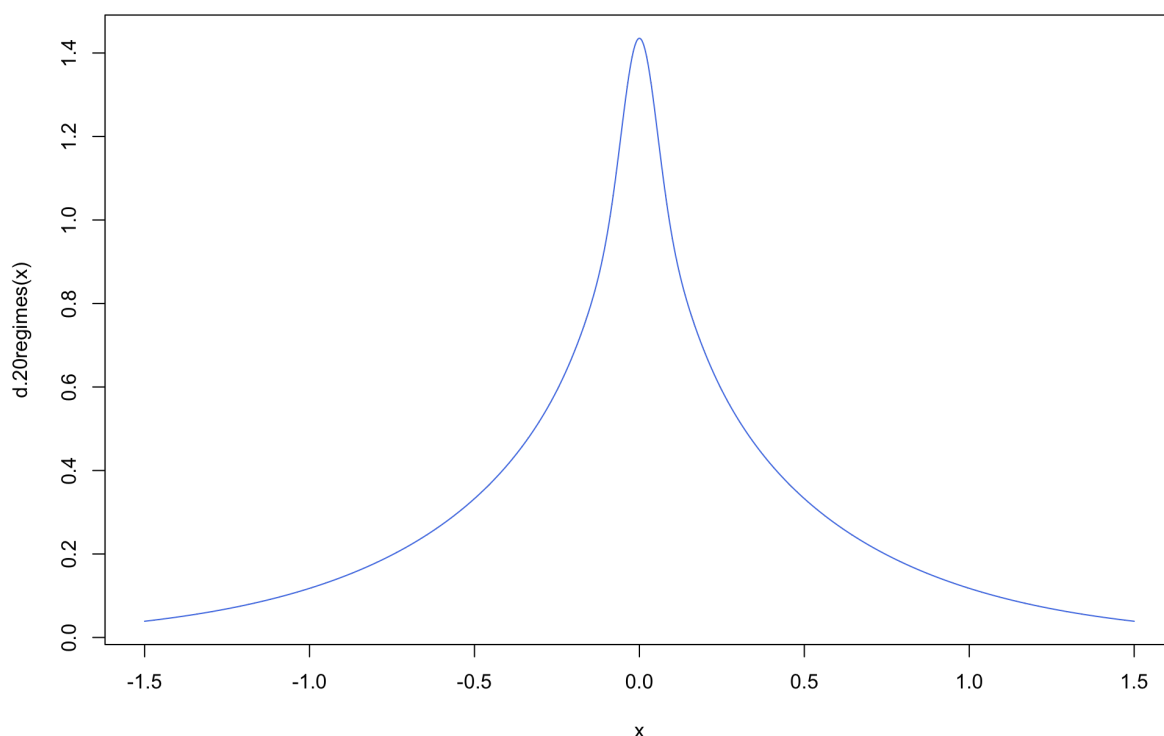


Figure 4: The 20-regime density in blue. Note the high central peak and fat tails.

Interpretation

- Fat tails and the high central peak are characteristics of mixtures of distributions with different variances.
 - This motivates us to model variance as a random variable.
- The volatility clustering feature implies that volatility (or variance) is auto-correlated.
 - In SV models, this is a consequence of the continuity of the volatility process.
- Note that simple jump-diffusion models do not have this volatility clustering property. After a jump, the stock price volatility does not change.
 - Self-exciting jump models can have the clustering property.

Economic argument for stationarity of the variance process

- Consider the distribution of the volatility of IBM in one hundred years time say.
- If the distribution of volatility were not stationary in the long term, the probability of the volatility of IBM being between 1% and 100% would be rather low.
- Since we believe that it is overwhelmingly likely that the volatility of IBM would in fact lie in that range, we deduce that the volatility process must be stationary in the

long term.

Power-law scaling of smiles and roughness

- [Alòs et al.]^[1] (2007), and later [Fukasawa]^[8] (2011), showed that a stochastic volatility model driven by fractional Brownian motion (fBm) reproduces power law scaling of smiles in the short-expiration limit.
- Ten years on, [Fukasawa]^[9] (2021) showed the converse: power-law scaling as $\tau^{H-\frac{1}{2}}$ forces volatility to be rough.
- All of this is inferred from option prices. For such a model to be realistic, volatility estimated from historical data should be rough too.
- We now follow [Volatility is Rough]^[11] and investigate the scaling properties of volatility time series.

The time series of realized variance

- We would like to study the time series of instantaneous variance V_t but of course cannot because V_t is latent.
- On the other hand, integrated variance $\frac{1}{\delta} \int_t^{t+\delta} V_s ds$ may (in principle) be estimated arbitrarily accurately given enough price data.
 - In practice, market microstructure noise makes estimation harder at very high frequency.
 - Sophisticated estimators of integrated variance have been developed to adjust for market microstructure noise. See Gatheral and Oomen^[14] (for example) for details of these.
- The Oxford-Man Institute of Quantitative Finance used to make historical realized variance (RV) estimates freely available.
 - Unfortunately, no longer. The last date in the dataset is 06/28/2022.
 - Each day, for 31 different indices, all trades and quotes were used to estimate realized (or integrated) variance over the trading day from open to close.
- Using daily RV estimates as proxies for instantaneous variance, we may investigate the time series properties of integrated variance empirically.
 - There are many different estimates of realized variance, all of them very similar. We will use the realized kernel estimates denoted by ".rk".

Load the data

```
In [8]: # Download .zip from GitHub
release_url <- "https://github.com/jgatheral/MTH9875/raw/main/9875-2.zip"
local_zip <- "9875-2.zip"

download.file(release_url, local_zip, mode = "wb")
unzip(local_zip)
```

```
In [9]: library(quantmod)
library(repr)
library(colorspace)
options(repr.plot.width=10, repr.plot.height=7, repr.plot.res=200)
```

```
In [10]: load("OxfordRV.rData")

names(rv.list)

'.AEX' · '.AORD' · '.BFX' · '.BSES' · '.BVLG' · '.BVSP' · '.DJI' · '.FCHI' · '.FTMIB' · '.FTSE' ·
'.GDAXI' · '.GSPTSE' · '.HSI' · '.IBEX' · '.IXIC' · '.KS11' · '.KSE' · '.MXX' · '.N225' · '.NSEI' ·
'.OMXC20' · '.OMXHPI' · '.OMXSPI' · '.OSEAX' · '.RUT' · '.SMSI' · '.SPX' · '.SSEC' · '.SSMI' ·
'.STI' · '.STOXX50E'
```

Set up nice colors

```
In [11]: my.col <- sequential_hcl(5, palette="Batlow")
bl <- "royalblue"
rd <- "red2"
pk <- "hotpink1"
gr <- "green4"
br <- "brown"
pu <- "purple"
or <- "orange"
```

Let's plot SPX realized variance.

```
In [12]: spx.rk <- rv.list[[".SPX"]]
stoxx.rk <- rv.list[[".STOXX50E"]]
ftse.rk <- rv.list[[".FTSE"]]
```

```
In [13]: plot(log(spx.rk), main="Log of SPX realized variance", col=rd)
```

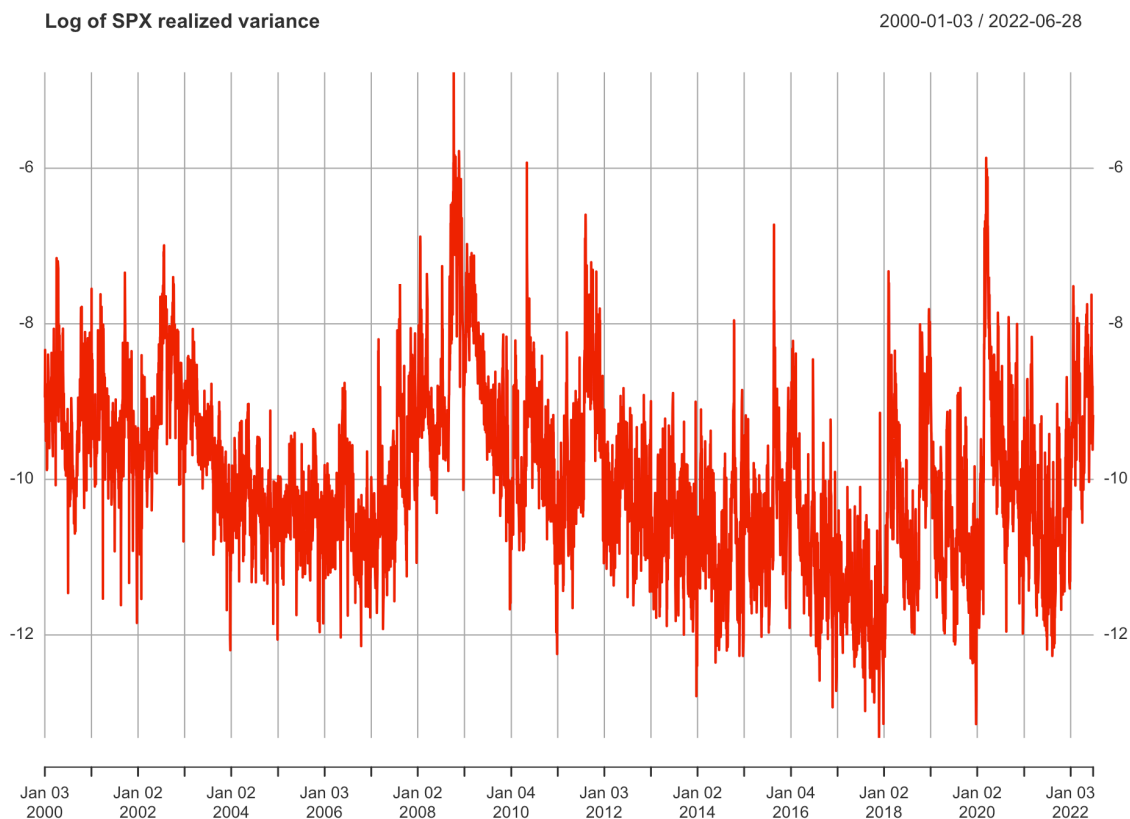


Figure 5: Oxford-Man Log KRV estimates of SPX realized variance from January 2000 to the current date.

```
In [14]: print(head(spx.rk))
         print(tail(spx.rk))
```

```
          [,1]
2000-01-03 1.301572e-04
2000-01-04 1.622259e-04
2000-01-05 2.398365e-04
2000-01-06 1.322324e-04
2000-01-07 9.486773e-05
2000-01-10 1.121113e-04
          [,1]
2022-06-17 2.234366e-04
2022-06-22 1.367900e-04
2022-06-23 1.499486e-04
2022-06-24 6.649679e-05
2022-06-27 9.746335e-05
2022-06-28 1.039309e-04
```

Scaling of the volatility process

For $q \geq 0$, we define the q th sample moment of differences of log-volatility at a given lag Δ . ($\langle \cdot \rangle$ denotes the sample average):

$$m(q, \Delta) = \langle |\log \sigma_{t+\Delta} - \log \sigma_t|^q \rangle$$

For example

$$m(2, \Delta) = \langle (\log \sigma_{t+\Delta} - \log \sigma_t)^2 \rangle$$

is just the sample variance of differences in log-volatility at the lag Δ .

Scaling of $m(q, \Delta)$ with lag Δ

```
In [15]: sig <- sqrt(as.numeric(spx.rk))

mq.del.Raw <- function(q, lag){mean(abs(diff(log(sig), lag=lag))^q)}
mq.del <- function(x, q){sapply(x, function(x){mq.del.Raw(q, x)})}

# Plot mq.del(1:100, q) for various q

x <- 1:100

mycol <- rainbow(5)

ylab <- expression(paste(log, " ", m(q, Delta)))
xlab <- expression(paste(log, " ", Delta))

qVec <- c(.5, 1, 1.5, 2, 3)
zeta.q <- numeric(5)
q <- qVec[1]

In [16]: options(repr.plot.height=7, repr.plot.width=10)

In [17]: plot(log(x), log(mq.del(x, q)), pch=20, cex=.5,
              ylab=ylab, xlab=xlab, ylim=c(-3, -.5))
fit.lm <- lm(log(mq.del(x, q)) ~ log(x))
abline(fit.lm, col=my.col[1], lwd=2)
zeta.q[1] <- coef(fit.lm)[2]

for (i in 2:5){
  q <- qVec[i]
  points(log(x), log(mq.del(x, q)), pch=20, cex=.5)
  fit.lm <- lm(log(mq.del(x, q)) ~ log(x))
  abline(fit.lm, col=my.col[i], lwd=2)
  zeta.q[i] <- coef(fit.lm)[2]
}
legend("bottomright", c("q = 0.5", "q = 1.0", "q = 1.5", "q = 2.0", "q = 3.0"),
      inset=0.05, lty=1, col = my.col)

print(zeta.q)
```

[1] 0.08377466 0.16687154 0.24956083 0.33209300 0.49728148

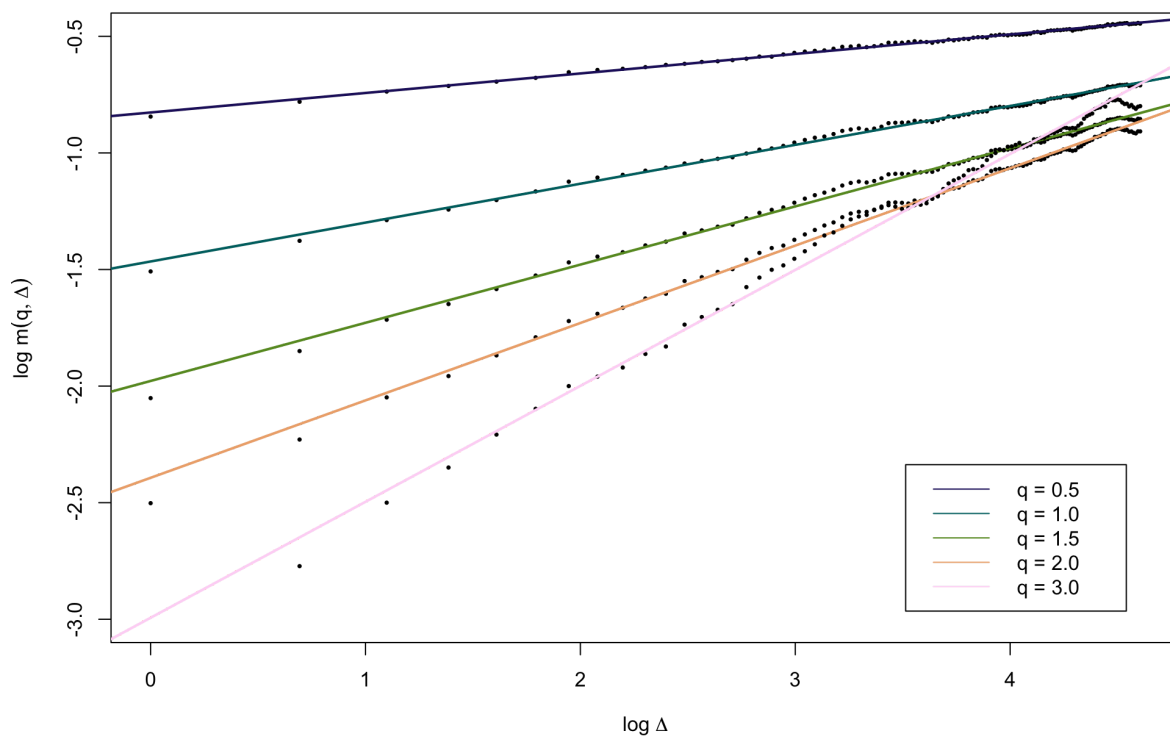


Figure 6: $\log m(q, \Delta)$ as a function of $\log \Delta$, SPX.

Monofractal scaling result

- From the above log-log plot, we see that for each q , $m(q, \Delta) \propto \Delta^{\zeta_q}$.
- How does ζ_q scale with q ?

Scaling of ζ_q with q

```
In [18]: plot(qVec, zeta.q, xlab="q", ylab=expression(zeta[q]), pch=20, col=bl, cex=2)
fit.lm <- lm(zeta.q[1:4] ~ qVec[1:4]+0)
abline(fit.lm, col=rd, lwd=2)
(h.est <- coef(fit.lm)[1])
```

qVec[1:4]: 0.16630481565354

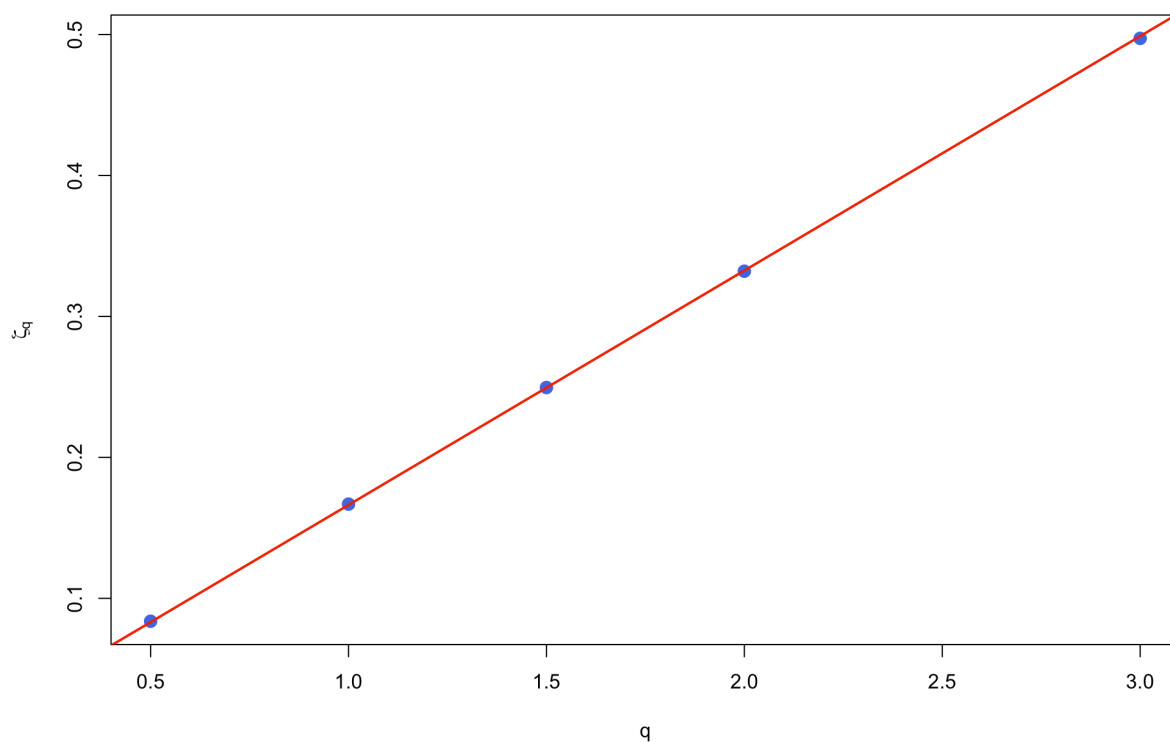


Figure 7: Scaling of ζ_q with q .

We find the monofractal scaling relationship

$$\zeta_q = q^H$$

with $H \approx 0.17$.

- Note however that H does vary over time, in a narrow range, as we will see later.
- Note also that our estimate of H is biased high because we proxied instantaneous variance V_t with its average over each day $\frac{1}{T} \int_0^T V_t dt$, where T is one trading day.
 - On the other hand, the time series of realized variance is noisy and this causes our estimate of H to be biased low.
- This scaling property as $\Delta \rightarrow 0$ is equivalent to H -Hölder continuity of paths of the volatility.
 - Since $H \ll 1/2$, *volatility is rough!*

Estimated H for all indices

We now repeat this analysis for all 31 indices in the Oxford-Man dataset.

```
In [19]: n <- length(rv.list)
h <- numeric(n) # H is estimated as half of the slope
nu <- numeric(n)

for (i in 1:n){ # Run all the regressions
  v <- rv.list[[i]]
  sig1 <- sqrt(abs(as.numeric(v)))

  xx <- 1:100
  dlsig2 <- function(lag){mean((diff(log(sig1),lag=lag))^2)}
  dlsig2Vec <- function(x){sapply(x,dlsig2)}

  fit.lm <- lm(log(dlsig2Vec(xx)) ~ log(xx))

  nu[i] <- sqrt(exp(coef(fit.lm)[1]))
  h[i] <- coef(fit.lm)[2]/2
}
```

```
In [20]: (OxfordH <- data.frame(names(rv.list),h.est=h,nu.est=nu))
```

A data.frame: 31 × 3

names.rv.list.	h.est	nu.est
<chr>	<dbl>	<dbl>
.AEX	0.15116126	0.2794888
.AORD	0.11051645	0.3037204
.BFX	0.14397740	0.2590815
.BSESN	0.13365442	0.2906722
.BVLG	0.14173291	0.2370132
.BVSP	0.13744512	0.2932294
.DJI	0.16790232	0.2838851
.FCHI	0.14176726	0.2870727
.FTMIB	0.14089962	0.2739568
.FTSE	0.13898212	0.2823674
.GDAXI	0.15540606	0.2670089
.GSPTSE	0.14469026	0.3000855
.HSI	0.12339313	0.2305324
.IBEX	0.13171773	0.2693158
.IXIC	0.15594794	0.2897655
.KS11	0.12479648	0.2773506
.KSE	0.11244817	0.3891245
.MXX	0.09265060	0.2851999
.N225	0.13216697	0.2993542
.NSEI	0.12837502	0.3185341
.OMXC20	0.11454790	0.2984794
.OMXHPI	0.12444437	0.3141314
.OMXSPI	0.13371935	0.3058484
.OSEAX	0.13926292	0.2614015
.RUT	0.12839201	0.3496840
.SMSI	0.11575621	0.3198488
.SPX	0.16604650	0.3022089
.SSEC	0.13577949	0.3194991
.SSMI	0.18768451	0.1886540
.STI	0.06749902	0.2394070

names.rv.list.	h.est	nu.est
<chr>	<dbl>	<dbl>
.STOXX50E	0.11488373	0.3543745

Distributions of $(\log \sigma_{t+\Delta} - \log \sigma_t)$ for various lags Δ

Having established these beautiful scaling results for the moments, how do the histograms look?

```
In [21]: plotScaling <- function(j,scaleFactor){
  v <- as.numeric(rv.list[[j]])
  x <- 1:100

  xDel <- function(x,lag){diff(x,lag=lag)}
  sd1 <- sd(xDel(log(v),1))
  sdL <- function(lag){sd(xDel(log(v),lag))}

  h <- OxfordH$h.est[j]

  plotLag <- function(lag){
    y <- xDel(log(v),lag)
    hist(y,breaks=100,freq=F,main=paste("Lag =", lag, "Days"),xlab=NA)# Very l
    curve(dnorm(x,mean=mean(y),sd=sd(y)),add=T,col=rd,lwd=2)
    curve(dnorm(x,mean=0,sd=sd1*lag^h),add=T,lty=2,lwd=2,col=bl)
  }

  (lags <- scaleFactor^(0:3))
  print(names(rv.list)[j])
  par(mfrow=c(2,2))
  par(mar=c(3,2,1,3))
  for (i in 1:4){plotLag(lags[i])}
  par(mfrow=c(1,1))
}
```

```
In [22]: options(repr.plot.height=5, repr.plot.width=10)
```

```
In [23]: plotScaling(27,5)
```

```
[1] ".SPX"
```

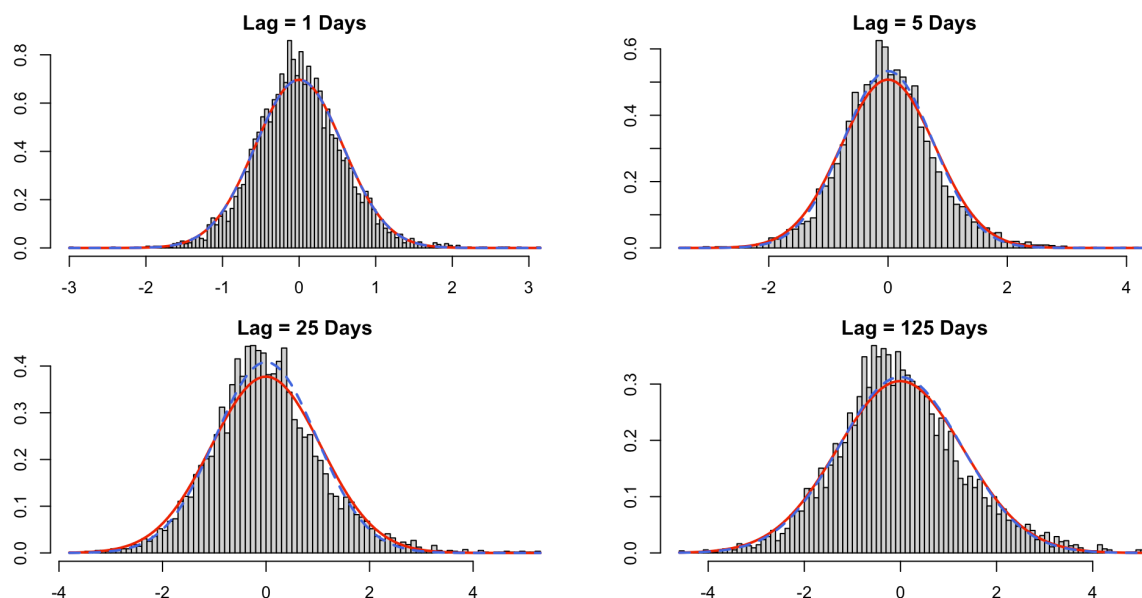



Figure 8: Histograms of $(\log \sigma_{t+\Delta} - \log \sigma_t)$ for various lags Δ ; normal fit in red; $\Delta = 1$ normal fit scaled by Δ^H in blue.

Universality?

- [Gatheral, Jaisson and Rosenbaum]^[13] compute daily realized variance estimates over one hour windows for DAX and Bund futures contracts, finding similar scaling relationships.
- We have also checked that Gold and Crude Oil futures scale similarly.
 - Although the increments $(\log \sigma_{t+\Delta} - \log \sigma_t)$ seem to be fatter tailed than Gaussian.
- [Bennedsen et al.]^[6], estimate volatility time series for more than five thousand individual US equities, finding rough volatility in every case.

A natural model of realized volatility

- Distributions of differences in the log of realized variance are close to Gaussian.
 - This motivates us to model $\Delta \log V_t$ as a normal random variable.
- Moreover, the scaling property of variance of RV differences suggests the model:

(1)

$$\log \sigma_{t+\Delta} - \log \sigma_t = \nu \left(W_{t+\Delta}^H - W_t^H \right)$$

where W^H is fractional Brownian motion.

- Indeed, if H is constant, (1) is the *unique* model consistent with Gaussianity of log differences, the observed scaling, and continuity of the volatility process.

Fractional Brownian motion (fBm)

- *Fractional Brownian motion* (fBm) $\{W_t^H; t \in \mathbb{R}\}$ is the unique Gaussian process with mean zero and autocovariance function

$$\mathbb{E}[W_t^H W_s^H] = \frac{1}{2} \left\{ |t|^{2H} + |s|^{2H} - |t-s|^{2H} \right\}$$

where $H \in (0, 1)$ is called the *Hurst index* or parameter.

- In particular, when $H = 1/2$, fBm is just Brownian motion.
- If $H > 1/2$, increments are positively correlated ("trending").
- If $H < 1/2$, increments are negatively correlated ("reverting").

Does simulated RSFV data look real?

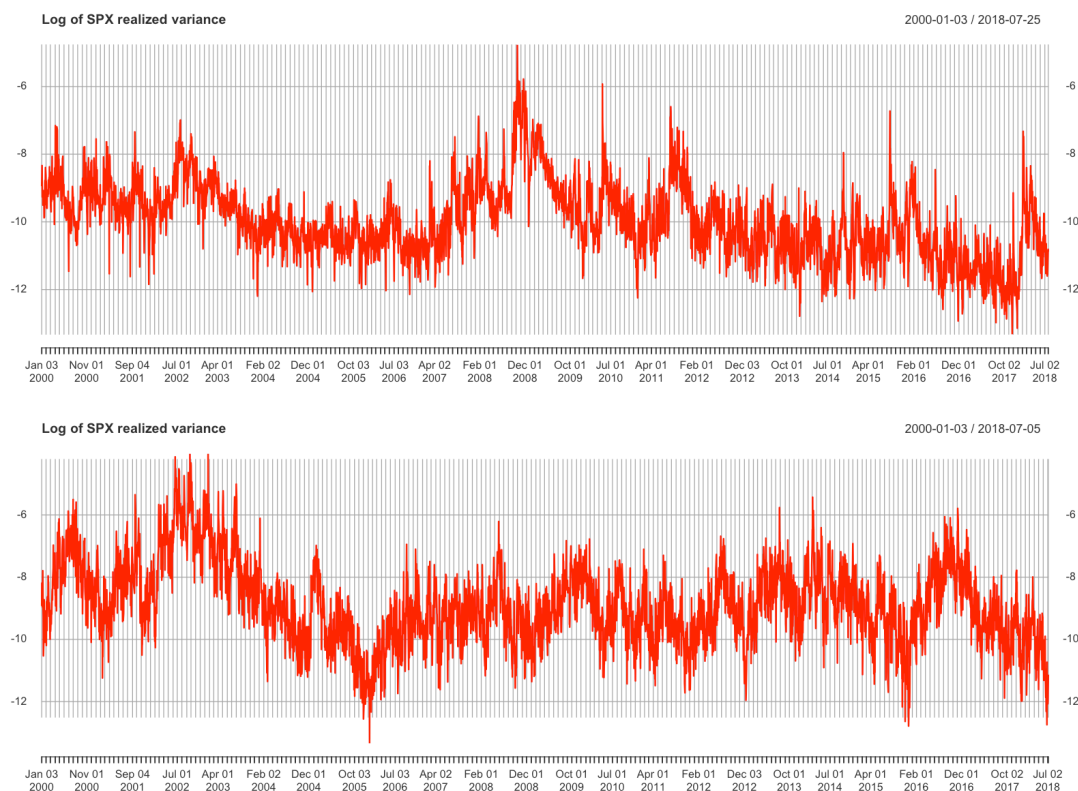


Figure 9: Volatility of SPX (above) and of the RSFV model (below).

Remarks on the comparison

- In respect of roughness, the simulated and actual graphs look very alike.
 - Persistent periods of high volatility alternate with low volatility periods.
- $H \sim 0.1$ generates very rough looking sample paths (compared with $H = 1/2$ for Brownian motion).
- Hence *rough volatility*.
- On closer inspection, we observe fractal-type behavior.
- The graph of volatility over a small time period looks like the same graph over a much longer time period.
- This feature of volatility has been investigated both empirically and theoretically in, for example, [Bacry and Muzy]^[3].
 - In particular, their Multifractal Random Walk (MRW) is related to a limiting case of the RSFV model as $H \rightarrow 0$.

Two estimates of H

- Now we have two estimates of the roughness of the volatility process:
 - $H_0 < H$ where H is from power-law scaling of the smile (under $\mathbb{Q}$).
 - $H_0 \sim 0, 1$ on average from realized variance data (under $\mathbb{P}$).
- Of course, H_0 should be the same under $\mathbb{P}$ and $\mathbb{Q}$.
 - $\mathbb{P}$ and $\mathbb{Q}$ are *equivalent*: they agree on which events have probability zero.
 - Hölder regularity is a property of the path itself. If $\log \sigma_t$ is H -Hölder $\mathbb{P}$ -a.s., it is H -Hölder $\mathbb{Q}$ -a.s.

Roughness is invariant under change of measure. The risk premium is not.

The stock price process

(2)

$$dS_t = \mu_t^{\mathbb{P}} S_t dt + \sqrt{V_t} S_t dZ_t^{\mathbb{P}}$$

- To ensure no-arbitrage, the stock price S is modeled as a positive semimartingale.
 - This excludes for example fractional Brownian motion with $H \neq \frac{1}{2}$.
 - If $H > 1/2$, quadratic variation (QV) vanishes.
 - If $H < 1/2$, QV is infinite!
 - The form (2) is completely general. Properties of the model depend on specific choices for μ_t and V_t .

A big simplifying assumption

- We will henceforth set interest rates and dividends to zero.
 - It is typically easy to reintroduce nonzero rates and dividends - but of course everything gets more complicated.
- With this assumption, the drift μ_t of the stock price process under $\mathbb{P}$ is the *equity risk premium* - the extra return that investors require for holding the risky stock.

$\mathbb{P}$ and $\mathbb{Q}$

- Philosophically, it's clear what the pricing measure $\mathbb{Q}$ is:
 - Probabilities are just prices associated with events.
- It's not clear at all (at least to me) what $\mathbb{P}$ is.
 - However, once we specify the change of measure $\frac{d\mathbb{P}}{d\mathbb{Q}}$, we can always get $\mathbb{P}$ from $\mathbb{Q}$.

$\frac{d\mathbb{P}}{d\mathbb{Q}}$ from log utility

- In order to fix $\frac{d\mathbb{P}}{d\mathbb{Q}}$, consider log utility, $U(W) = \log W$.
- We assume that the optimal investment strategy is to keep some proportion π_t of wealth in stock at time t .
- What policy π maximizes $\log W_T$, the utility of terminal wealth W_T ?
- Under the dynamics (2),

$$dW_t = \pi_t W_t \frac{dS_t}{S_t} = \pi_t W_t \left\{ \mu_t dt + \sqrt{V_t} dZ_t^{\mathbb{P}} \right\}.$$

- Applying Itô's Formula,

$$\begin{aligned}
\log W_T &= \log W_t + \int_t^T d \log W_u \\
&= \log W_t + \int_t^T \frac{dW_u}{W_u} - \frac{1}{2} \int_t^T \frac{d\langle W \rangle_u}{W_u^2} \\
&= \log W_t + \int_t^T \pi_u \mu_u du + \int_t^T \pi_u \sqrt{V_u} dZ_u \\
&\quad - \frac{1}{2} \int_t^T \pi_u^2 V_u du.
\end{aligned}$$

- It follows that utility is maximized *pathwise* if $\pi_u = \mu_u / V_u$.
- If $\mu_s = \beta V_s$ for some constant $\beta > 0$, it is optimal to keep a constant proportion of wealth in stock.
- With this choice, the *equity risk premium*, the extra return that investors require for taking on risk, is proportional to instantaneous variance.

Change of measure

- The dynamics (2) are under $\mathbb{P}$.
 - Recall that according to the Fundamental Theorem of Asset Pricing, S_t must be a martingale under the pricing measure $\mathbb{Q}$.
- We now assume that $\mu_s = \beta V_s$ and define a change of measure via the Girsanov transformation

$$dZ_t^{\mathbb{Q}} = dZ_t^{\mathbb{P}} + \beta \sqrt{V_t} dt.$$

- Benchmark pricing ([Platen]^[12]) corresponds to the special case $\beta = 1$.
- Then

$$dS_t = \sqrt{V_t} S_t dZ_t^{\mathbb{Q}}.$$

- In other words, in order for S to be a $\mathbb{Q}$ -martingale, we must have $\mu_u = \beta V_u$.

Modeling the variance process: Black-Scholes

$V_t = \sigma(t)^2$, a deterministic function of time.

Modeling the variance process: Classical Heston

The classical Heston variance process may be written as

$$dV_t = -\lambda^{\mathbb{P}} (V_t - \bar{V}^{\mathbb{P}}) dt + \eta \sqrt{V_t} dW_t^{\mathbb{P}}$$

with $d\langle W, Z \rangle_t = \rho dt$.

- $\lambda^{\mathbb{P}}$ is the speed of reversion of V_t to its long term mean $\bar{V}^{\mathbb{P}}$
- η is the volatility of volatility.

Heston change of measure

Making the analogous change of measure (with $\beta = 1$),

$$dW_t^{\mathbb{Q}} = dW_t^{\mathbb{P}} + \rho \sqrt{V_t} dt,$$

we get

$$dV_t = -\lambda^{\mathbb{Q}} (V_t - \bar{V}^{\mathbb{Q}}) dt + \eta \sqrt{V_t} dW_t^{\mathbb{Q}},$$

where

$$\lambda^{\mathbb{Q}} = \lambda^{\mathbb{P}} + \rho \eta; \quad \lambda^{\mathbb{Q}} \bar{V}^{\mathbb{Q}} = \lambda^{\mathbb{P}} \bar{V}^{\mathbb{P}}.$$

We see that the process under $\mathbb{Q}$ has the same form as the process under $\mathbb{P}$.

Who is interested in $\mathbb{P}$ vs $\mathbb{Q}$?

- Anybody who wants to predict option prices should be interested in the relationship between $\mathbb{P}$ and $\mathbb{Q}$:
 - Market makers
 - Volatility traders
- On the other hand, exotic option bookrunners are not at all interested.
 - Everything they do is under $\mathbb{Q}$!
- Option pricing models are all under $\mathbb{Q}$.

$\mathbb{Q}$ from option prices

- In practical applications of SV models, we always assume that the SDEs for S and V are under the pricing measure $\mathbb{Q}$ because we are invariably interested in fitting models to option prices.
 - We effectively impute $\mathbb{Q}$ by fitting the parameters of the process we are imposing.

- From now on, we will drop the superscripts $\mathbb{P}$ and $\mathbb{Q}$.
 - Unless otherwise stated, everything will be under $\mathbb{Q}$.

Forward variance models

- In both examples, Black-Scholes and classical Heston, the volatility process is modeled directly.
 - However V_t is not observable.
- Bergomi and Guyon^[3], suggested that it makes more sense to model forward variances

$$\xi_t(u) = \mathbb{E}_t^{\mathbb{Q}}[V_u], \quad u > t.$$

- The forward variances, being conditional expectations under $\mathbb{Q}$, are tradable.
 - Not only in principle, but in practice, as we will see later in the lecture series.
- The forward variance processes are modeled (in the single-factor case) as

$$d\xi_t(u) = \eta_t(u; \omega) dW_t,$$

where as before $d\langle W, Z \rangle_t = \rho dt$.

- The $\mathbb{R}_{\geq 0}$ -valued stochastic process $\eta_t(u; \omega)$ is progressively measurable for all $u > 0$.
 - Conventionally, it is assumed that η is adapted to the filtration generated by W . In other words, the variance process depends only on the history of the variance process.
 - A truly path-dependent model would have η adapted to the filtration jointly generated by W and Z .
- If V is continuous and uniformly integrable, we can recover V_t from $\xi_t(u)$ as $V_t = \lim_{u \downarrow t} \xi_t(u)$.
 - For our purposes, $V_t = \xi_t(t)$.
- The initial conditions of a typical forward variance model are the initial stock price S_t and the initial forward variance curve $\xi_t(u)_{u>t}$.

The Black-Scholes-Merton (BSM) PDE

- Consider the value of some claim C on S .
 - There is only one stochastic process S , so C can only be a function of t and S .
 - For example, C could be a European call, in which case

$$C(t, S_t) = \mathbb{E}_t [(S_T - K)^+]$$

- Applying Itô's Formula,

$$\begin{aligned} dC &= C_t dt + C_S dS_t + \frac{1}{2} C_{SS} d\langle S \rangle_t \\ &= C_t dt + C_S dS_t + \frac{1}{2} \sigma(t)^2 S_t^2 C_{SS} dt. \end{aligned}$$

- Now C is priced as the conditional expectation of the payoff under the pricing measure, which is a $\mathbb{Q}$ -martingale. So

$$\mathbb{E}_t [dC] = 0.$$

- Using that $\mathbb{E}_t [dS_t] = 0$, we get

$$\frac{dC}{dt} + \frac{1}{2} \sigma(t)^2 S_t^2 \frac{\partial^2 C_t}{\partial S^2} = 0,$$

the familiar BSM equation, $\theta + \gamma = 0$.

The classical Heston PDE

- Now there are two stochastic processes, S and V . So C is a function of t , S and V .
- Applying Itô's Formula again,

$$\begin{aligned} dC &= C_t dt + C_S dS_t + C_V dV_t + \frac{1}{2} C_{SS} d\langle S \rangle_t \\ &\quad + C_{SV} d\langle S, V \rangle_t + \frac{1}{2} C_{VV} d\langle V \rangle_t \\ &= C_t dt + C_S dS_t - \frac{\lambda}{2} (V_t - \bar{V}) C_V dt + \eta \sqrt{V_t} C_V dW_t \\ &\quad + \frac{1}{2} V_t S_t^2 C_{SS} dt \end{aligned}$$

- $\rho \eta \sqrt{V_t} C_{SV} dt + \frac{1}{2} \eta^2 V_t C_{VV} dt$.

- Taking the conditional expectation gives the classical Heston equation

$$\begin{aligned} \frac{dC}{dt} + \frac{1}{2} V_t S_t^2 \frac{\partial^2 C}{\partial S^2} - \lambda (V_t - \bar{V}) \frac{\partial C}{\partial V} \\ + \rho \eta S_t V_t \frac{\partial^2 C}{\partial S \partial V} + \frac{1}{2} \eta^2 V_t \frac{\partial^2 C}{\partial V^2} = 0. \end{aligned}$$

- In words,

$$\theta + \gamma + \nu + \nu\nu + \nu\nu = 0.$$

Extension to n factors

- In a (calibrated) one factor SV model, given the stock price S and the market price of *only one option*, in principle we know the prices of all options.
 - This is obviously extremely unrealistic.
- It may be desirable (essential if jointly calibrating to the prices of volatility derivatives) to include more factors. For example, in the Double Mean Reverting (DMR) model [Bayer, Gatheral and Karlsmark]^[4]:

$$\begin{aligned} dS &= \sqrt{V^{(1)}_t} S dZ^{(1)}_t \\ dV^{(1)}_t &= -\kappa_1 (V^{(1)}_t - V^{(2)}_t) dt + \eta_1 V^{(1)}_t^{\alpha_1} dZ^{(2)}_t \\ dV^{(2)}_t &= -\kappa_2 (V^{(2)}_t - \bar{V}) dt + \eta_2 V^{(2)}_t^{\alpha_2} dZ^{(3)}_t \end{aligned}$$

with $d\langle Z^{(i)}, Z^{(j)} \rangle_t = \rho_{ij} dt$.

- In the DMR model, we know all option prices given the stock price and the prices of two options.
- In some general n -factor model, we would know all option prices given the prices of n options.

Market completeness

- Academics often say that SV models are not complete. This is true only if there are no traded derivative securities.
- In reality, stochastic volatility models cannot be calibrated to fit the prices of all options simultaneously.
 - So, on the contrary, there are insufficient factors to fit all options.

Solving the valuation PDEs

- How easy these PDEs are to solve depends on the model and the boundary conditions.
- Fast solutions for European options are needed for calibration to market option prices.
 - The boundary condition for a European call option is $C(S_T, T) = (S_T - K)^+$.
- For general payoffs (i.e. boundary conditions), we solve the valuation equation using numerical PDE or Monte Carlo techniques.
 - This is too slow for efficient calibration.
- Consequently, models for which fast exact or approximate European option solutions exist are the most popular.

Local Volatility

Local volatility: Historical development

- Stochastic volatility models are hard to handle
 - difficult to compute with and hard (until recently impossible) to calibrate.
- Practitioners need a simple model that prices European options consistently with the market.

Risk-neutral density from prices

The marginal risk-neutral density may be recovered by twice differentiating European call prices wrt strike: $p(K) = \frac{\partial^2 C}{\partial K^2}$.

Proof.

$$C(K) = \mathbb{E}[(S_T - K)^+]$$

so

$$-\frac{\partial C}{\partial K} = \mathbb{E}[\theta(S_T - K)] = \Pr(S_T > K)$$

and

$$\frac{\partial^2 C}{\partial K^2} = \mathbb{E}[\delta(S_T - K)] = p(K).$$

The trader version: The butterfly

- Consider the payoff of the following option combination:
 - Long a call struck at $K + \Delta K$,
 - short 2 calls struck at K ,
 - long a call struck at $K - \Delta K$.

```
In [24]: callPayoff <- function(x,K){pmax(x-K,0)}
putPayoff <- function(x,K){pmax(K-x,0)}

butterfly <- function(x){callPayoff(x,.99)-2*callPayoff(x,1)+callPayoff(x,1.1)}
curve(butterfly(x),from=.8,to=1.2,ylim=c(-0.005,0.015),col="red",xlab="Strike",ylab="Payoff")
```

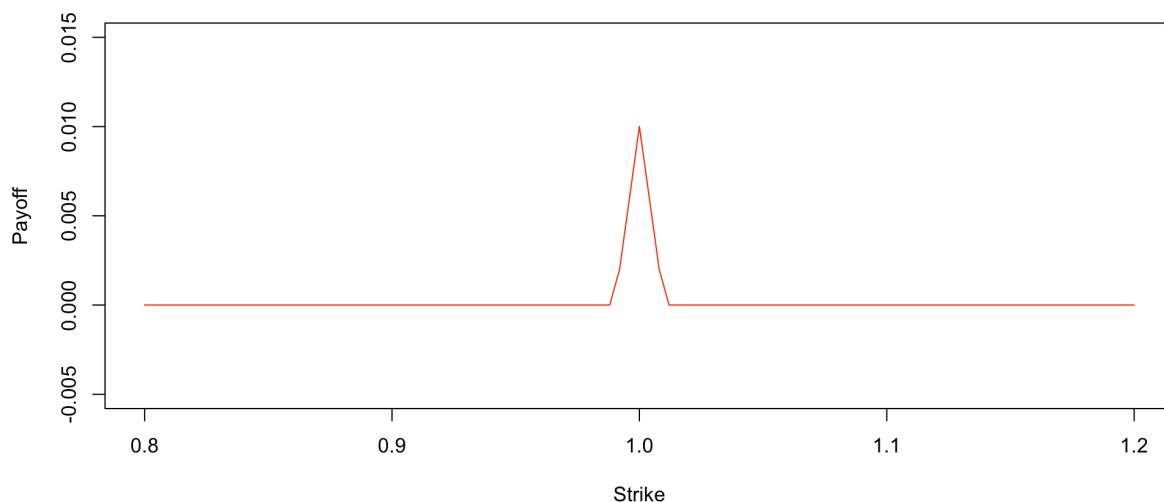


Figure 10: Payoff of a butterfly with strikes $\{0.99, 1.00, 1.01\}$.

The key idea

- For as long as options have been traded (probably), it has been understood that the risk-neutral density could be derived from the market prices of European options.
 - This is the famous Breeden-Litzenberger result.
- The breakthrough came when [Dupire]^[7] and [Derman and Kani]^[5] noted that under risk-neutrality, there was a unique diffusion process consistent with these distributions.
 - The corresponding unique state-dependent diffusion coefficient $\sigma_\ell(S, t)$ consistent with current European option prices is known as the *local volatility*.

function.

Local volatility: Philosophy

- It is unlikely that Dupire, Derman and Kani (DDK) ever thought of local volatility as representing a model of how volatilities actually evolve.
 - [Dumas, Fleming and Whaley]^[6] confirmed empirically that the dynamics of the implied volatility surface were not consistent with the assumption of constant local volatilities.
- They likely thought of local volatilities as representing some kind of average over all possible instantaneous volatilities in a stochastic volatility world (an “effective theory”).
 - We will prove a precise version of this idea later in this lecture, formalizing the DDK intuition.
- Local volatility is the simplest extension of Black-Scholes that allows practitioners to price exotic options consistently with the known prices of vanilla options.

A Brief Review of Dupire’s Work

- Given T -expiration European call prices $\{C(S_0, K, T)\}$, the risk neutral density function φ of the final spot S_T is given by

$$\varphi(K, T; S_0) = \frac{\partial^2 C}{\partial K^2}.$$

- Dupire showed that there is a unique risk-neutral diffusion process which generates a given marginal distribution of final spot prices S_T for each time T , conditional on some starting spot price S_0 .
- Equivalently, given the set of all European option prices, we may determine the functional form of the diffusion parameter (local volatility) of the unique risk neutral diffusion process which generates these prices.

A simplification

For simplicity, in what follows, we will assume zero interest rates and zero dividends.

Local volatility dynamic - the Dupire equation

The resulting *local volatility* model reads

$$\frac{dS_t}{S_t} = \sigma_\ell(S_t, t) dZ_t.$$

The local volatility function $\sigma_\ell(\cdot)$ is given by the Dupire equation:

The Dupire equation

(3)

$$\sigma_\ell^2(K, T) = \frac{\frac{\partial C}{\partial T}}{\frac{1}{2} K^2 \frac{\partial^2 C}{\partial K^2}} \quad (1)$$

- The right hand side of equation (3) can in principle be computed from known European option prices.
 - Given a complete set of European option prices for all strikes and expirations, local volatilities are given uniquely by equation (2).
 - In words, "local variance is given by the ratio of a calendar spread to a butterfly".
- We can view equation (3) as a *definition* of the local volatility function regardless of what kind of process (stochastic volatility for example) actually governs the evolution of volatility.

Local variance as a conditional expectation of instantaneous variance

Following Derman and Kani, write

$$C(S_0, K, T) = \mathbb{E}[(S_T - K)^+]$$

As before, differentiating twice with respect to K gives

$$\frac{\partial^2 C}{\partial K^2} = \mathbb{E}[\delta(S_T - K)]$$

where $\delta(\cdot)$ is the Dirac δ function. Now, a formal application of Itô's Lemma to the terminal payoff of the option gives

(4)

$$d(S_T - K)^+ = \theta(S_T - K) dS_T + \frac{1}{2} V_T S_T^2 \delta(S_T - K) dT$$

Taking conditional expectations of each side, and using the fact that S_T is a martingale, we get

$$dC = d\mathbb{E}[(S_T - K)^+] = \frac{1}{2} \mathbb{E}[V_T S_T^2 \delta(S_T - K)] dT$$

Also, denoting the final density by $p(S_T, T)$, we can write

$$\begin{aligned} \mathbb{E}[V_T S_T^2 \delta(S_T - K)] &= \mathbb{E}[V_T | S_T = K] K^2 \mathbb{E}[\delta(S_T - K)] \\ &= \mathbb{E}[V_T | S_T = K] K^2 p(K, T) \\ &= \mathbb{E}[V_T | S_T = K] K^2 \frac{\partial^2 C}{\partial K^2} \end{aligned}$$

Putting this together, we get

(5)

$$\frac{\partial C}{\partial T} = \mathbb{E}[V_T | S_T = K] \frac{1}{2} K^2 \frac{\partial^2 C}{\partial K^2}$$

Comparing (4) with the definition (2) of local volatility, we get

Local variance as conditional expectation

$$\sigma_\ell^2(K, T) = \mathbb{E}[V_T | S_T = K]$$

That is, local variance is the risk-neutral expectation of the instantaneous variance conditional on the final stock price S_T being equal to the strike price K .

Tanaka's formula

Sometimes in the literature, we see (4) written in integral form as

$$\begin{aligned} (S_T - K)^+ &= (S_0 - K)^+ + \int_0^T \mathbf{1}_{S_t > K} dS_t + \frac{1}{2} \int_0^T V_t S_t^2 \delta(S_t - K) dt \end{aligned}$$

This is a form of Tanaka's formula. It expresses the payoff of a European call as the proceeds of a stop-loss start-gain hedging strategy plus local time.

Local volatility in terms of implied volatility

Market prices of options are quoted in terms of Black-Scholes implied volatility $\Sigma(K, T; S_0)$. In other words, we may write

$$C(S_0, K, T) = C_{BS}(S_0, K, \Sigma(K, T), T)$$

It will be more convenient for us to work in terms of two dimensionless variables: the Black-Scholes *implied total variance* w defined by

$$w(K, T) := \Sigma^2(K, T) T$$

and the log-strike k defined by

$$k = \log\left(\frac{K}{S_t}\right).$$

- We are again assuming zero rates and dividends here so that the forward price $\mathbb{E}_t[S_T] = S_t$.

Black-Scholes formula in terms of log-strike and total variance

In terms of these variables, the Black-Scholes formula for the undiscounted call price becomes

(6)

$$\begin{aligned} C_{BS} &= S_t \{N(d_+) - e^k N(d_-)\} \\ &= S_t \left\{ N\left(-\frac{k}{\sqrt{w}} + \frac{\sqrt{w}}{2}\right) - e^k N\left(-\frac{k}{\sqrt{w}} - \frac{\sqrt{w}}{2}\right) \right\} \end{aligned}$$

and the Dupire equation (3) becomes

(7)

$$\frac{\partial C}{\partial T} = \frac{v_\ell}{2} \left\{ \frac{\partial^2 C}{\partial k^2} - \frac{\partial C}{\partial k} \right\}$$

with $v_\ell = \sigma_\ell^2(K, T)$ representing the local variance.

Now, by taking derivatives of the Black-Scholes formula (5), we obtain

(8)

$$\begin{aligned}
\frac{\partial^2 C_{BS}}{\partial w^2} &= \left(-\frac{1}{8} - \frac{1}{2w} + \frac{k^2}{2w^2} \right) \frac{\partial C_{BS}}{\partial w} \\
\frac{\partial^2 C_{BS}}{\partial k \partial w} &= \left(\frac{1}{2} - \frac{k}{w} \right) \frac{\partial C_{BS}}{\partial w} \\
\frac{\partial^2 C_{BS}}{\partial k^2} - \frac{\partial C_{BS}}{\partial k} &= 2 \frac{\partial C_{BS}}{\partial w}
\end{aligned} \tag{2}$$

Dupire equation in terms of implied variance

We may transform equation (7) into an equation in terms of implied variance by making the substitutions

$$\begin{aligned}
\frac{\partial C}{\partial k} &= \frac{\partial C_{BS}}{\partial k} + \frac{\partial C_{BS}}{\partial w} \frac{\partial w}{\partial k} \\
\frac{\partial^2 C}{\partial k^2} &= \frac{\partial^2 C_{BS}}{\partial k^2} + 2 \frac{\partial^2 C_{BS}}{\partial k \partial w} \frac{\partial w}{\partial k} + \frac{\partial^2 C_{BS}}{\partial w^2} \left(\frac{\partial w}{\partial k} \right)^2 + \frac{\partial C_{BS}}{\partial w} \frac{\partial^2 w}{\partial k^2} \\
\frac{\partial C}{\partial T} &= \frac{\partial C_{BS}}{\partial T} + \frac{\partial C_{BS}}{\partial w} \frac{\partial w}{\partial T} = \frac{\partial C_{BS}}{\partial w} \frac{\partial w}{\partial T},
\end{aligned}$$

where the last equality follows from the fact that the only explicit dependence of the option price on T in equation (5) is through the total implied variance w .

Using (8), the Dupire equation (3) now becomes

$$\begin{aligned}
&\frac{\partial C_{BS}}{\partial w} \frac{\partial w}{\partial T} \\
&= \frac{v_\ell}{2} \left\{ -\frac{\partial C_{BS}}{\partial k} + \frac{\partial^2 C_{BS}}{\partial k^2} - \frac{\partial C_{BS}}{\partial w} \frac{\partial w}{\partial k} + 2 \frac{\partial^2 C_{BS}}{\partial k \partial w} \frac{\partial w}{\partial k} + \right. \\
&\quad \left. \frac{\partial^2 C_{BS}}{\partial w^2} \left(\frac{\partial w}{\partial k} \right)^2 + \frac{\partial C_{BS}}{\partial w} \frac{\partial^2 w}{\partial k^2} \right\} \\
&= \frac{v_\ell}{2} \frac{\partial C_{BS}}{\partial w} \left\{ 2 - \frac{\partial w}{\partial k} + 2 \left(\frac{1}{2} - \frac{k}{w} \right) \frac{\partial w}{\partial k} \right. \\
&\quad \left. + \left(-\frac{1}{8} - \frac{1}{2w} + \frac{k^2}{2w^2} \right) \left(\frac{\partial w}{\partial k} \right)^2 + \frac{\partial^2 w}{\partial k^2} \right\}
\end{aligned}$$

Local variance in terms of implied variance

Then, taking out a factor of $\frac{\partial C_{BS}}{\partial w}$ and simplifying, we get

$$\frac{\partial w}{\partial T} = v_\ell \left\{ 1 - \frac{k}{w} \frac{\partial w}{\partial k} + \frac{1}{4} \left(-\frac{1}{4} - \frac{1}{w} + \frac{k^2}{w^2} \right) \left(\frac{\partial w}{\partial k} \right)^2 + \frac{1}{2} \frac{\partial^2 w}{\partial k^2} \right\}$$

Inverting this and rearranging slightly gives our final result (formula (1.10) of [TVS]^[10]):

Local volatility from implied volatility

(9)

$$v_\ell(k, T) = \frac{\frac{\partial w}{\partial T}}{\left(1 - \frac{1}{2} \frac{k}{w} \frac{\partial w}{\partial k} \right)^2 - \frac{1}{4} \left(\frac{1}{4} + \frac{1}{w} \right) \left(\frac{\partial w}{\partial k} \right)^2 + \frac{1}{2} \frac{\partial^2 w}{\partial k^2}} \quad (3)$$

Special Case: No Skew

If w is a function of T only and not of k (so no skew and no smile), then we must have

$$v_\ell = \frac{\partial w}{\partial T}$$

and the local variance reduces to the forward Black-Scholes implied variance. The solution to this is of course

$$w(T) = \int_0^T v_\ell(t) dt$$

The Berestycki-Busca-Florent (BBF) formula

Wlog, set $t = 0$. Following [Berestycki, Busca and Florent]^[3], suppose the Black-Scholes implied volatility $\Sigma(k, T) = \Sigma_0(k) + \mathcal{O}(T)$. Then, to zeroth order in T , formula (9) reads:

(10)

$$\sigma_\ell(k, T)^2 =: v_\ell(k, T) = \frac{\Sigma_0^2(k) + \mathcal{O}(T)}{\left(1 - \frac{k}{2w} \partial_k w \right)^2 + \mathcal{O}(T)} =: \sigma_\ell(k)^2. \quad (4)$$

Noting that

$$1 - \frac{k}{2w} \partial_k w = \sqrt{w} \frac{\partial}{\partial k} \left(\frac{k}{\sqrt{w}} \right) = \Sigma_0(k) \frac{\partial}{\partial k} \left(\frac{k}{\Sigma_0(k)} \right),$$

we may rearrange (9) to give (to zeroth order in T)

$$\frac{\partial}{\partial k} \left(\frac{k}{\Sigma_0(k)} \right) = \frac{1}{\sigma_\ell(k)}.$$

Integrating, we arrive at the solution:

The BBF formula

$$\frac{1}{\Sigma_0(k)} = \frac{1}{k} \int_0^k \frac{dz}{\sigma_\ell(z)} = \int_0^1 \frac{d\alpha}{\sigma_\ell(\alpha k)}.$$

Scaling of volatility smiles in practice

- As we say in Lecture 1, volatility smiles explode roughly as $1/\sqrt{\tau}$ as time to expiration $\tau := T - t \rightarrow 0$.
- The assumption $\sigma_{BS}(k, T) = \sigma_0(k) + \mathcal{O}(T)$ for short dates is not realistic.
 - Though this does describe the behavior of smiles generated by conventional stochastic volatility models.

The heat-kernel most-likely-path approximation

In a later paper, [Gatheral and Wang]^[8] derived the following fancier approximation:

$$\frac{k^2}{\Sigma^2(k, T)} \approx T \int_0^T \left[\frac{\dot{x}(t)}{\sigma_\ell(x(t), t)} \right]^2 dt$$

where the *variational most likely path* (VMLP) $x(t)$ is the solution of the following variational problem:

$$\begin{cases} -\frac{d}{dt} \left[\frac{\dot{x}(t)}{\sigma_\ell(x(t), t)} \right] + \frac{\partial_t \sigma_\ell(x, t)}{\sigma_\ell^2(x, t)} \dot{x}(t) = 0, \\ x(0) = 0, \quad x(T) = k. \end{cases}$$

This formula reduces to the BBF formula when the local volatility function $\sigma_\ell(x, t)$ is time-homogeneous and is much more accurate otherwise.

Equivalence to BBF in the time homogeneous case

Since $\partial_t \sigma_\ell(x, t) = 0$, the VMLP satisfies

$$\frac{d}{dt} \left[\frac{\dot{x}(t)}{\sigma_\ell(x)} \right] = 0.$$

Let

$$\phi(x) = \int_0^x \frac{d\xi}{\sigma_\ell(\xi)}.$$

Then

$$\frac{d\phi}{dt} = \frac{\dot{x}(t)}{\sigma_\ell(x)}$$

and

$$\frac{d^2\phi}{dt^2} = 0.$$

Then

$$\frac{d\phi}{dt} = \frac{\dot{x}(t)}{\sigma_\ell(x, t)} = \text{const.} = \frac{\phi(k)}{T}.$$

Thus

$$\frac{k^2}{\Sigma^2(k, T)} \approx T \int_0^T \left[\frac{\dot{x}(t)}{\sigma_\ell(x(t), t)} \right]^2 dt = T \int_0^T \left[\frac{\phi(k)}{T} \right]^2 dt = \phi(k)^2.$$

It follows that

$$\frac{1}{\Sigma(k, T)} \approx \frac{1}{k} \int_0^k \frac{d\xi}{\sigma_\ell(\xi)}.$$

Conclusion

- The heat-kernel approximation is the best bet in practice.
 - However, as shown in our paper, accuracy is not great.
- In practice, numerical PDE should be used to get the implied volatility surface given a local volatility function.

Local stochastic volatility (LSV) models

- Local volatility models fit the volatility surface perfectly, by construction.
 - But their dynamics are completely unrealistic, as we shall see.
- Stochastic volatility models don't fit observed option prices.
- Instead, sophisticated practitioners use local stochastic volatility models of the form

$$\frac{dS_t}{S_t} = A(S_t, t) \sqrt{V_t} dZ_t.$$

where V_t is modeled using stochastic volatility.

- $A(S_t, t)$ is sometimes called the *leverage* function.
 - Note in particular, that a better model for V should give a leverage function A closer to one.
- LSV models fit observed option prices perfectly, by construction.
 - And they have reasonable dynamics.
 - Pricing is with Monte Carlo.
- But they are very hard to calibrate.
 - Calibration is typically done in overnight batch mode.
- For a very nice more detailed explanation, see Chapter 9 of [Austing]^[2].

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